Response to Reviewers — YOLOv8-FLEO (MDPI Algorithms)

We thank the editor and all three reviewers for their careful and constructive reading of the manuscript. Below we address every comment point by point. Text in [brackets] marks values to be filled from the revised experiments in progress.

Reviewer 1

Comment 1.1: The rationale for setting subspace dimension d=8 is not provided. Please add ablation experiments evaluating algorithm accuracy and hardware cost under different d values to justify the selection of d=8.

Author Response: We thank the reviewer for the comment. We selected d=8 as the knee of the accuracy–cost trade-off. We have added an ablation over the per-emotion subspace width $d \in \{4, 8, 16, 32\}$ on RAF-DB, reporting both recognition accuracy and hardware cost. Because FLEO projects each of the two neck sites (P3, P4) to $K \cdot d = 7d$ channels, its parameter and MAC cost grow approximately linearly in d; relative to d=8 the FLEO footprint is 0.50× at d=4, 2.05× at d=16, and 4.30× at d=32. Accuracy [improves from d=4 to d=8 and then saturates: d=4 [**], d=8 [**], d=16 [**], d=32 [**] mAP50], so d=8 captures the representational benefit of the orthogonal subspaces at minimal accelerator footprint.

Author Action: We have added Table [X] and a subsection in Section 4 reporting the d-ablation (accuracy and hardware cost), and we have added one sentence to Section 3.1.2 stating the rationale for d=8. The training script `scripts/train.py` exposes the width via `--d` for full reproducibility.

Comment 1.2: Both RAF-DB and FER2013 are standard single-label benchmarks. How is the fuzzy label vector μ derived from annotator votes as described? Is it based on multi-annotator metadata from the datasets, or are soft labels generated algorithmically?

Author Response: We thank the reviewer for the comment. The soft labels are generated algorithmically; we do not use multi-annotator metadata. Equation (2) is the general membership definition in which n_c is the vote count for class c. For a single-label corpus this degenerates to $n_c = \mathbb{1}[c = y]$, so Equation (2) reduces to the additively-smoothed soft target $\mu_c = (\mathbb{1}[c=y] + \alpha \cdot \pi_c)/(1 + \alpha)$, computed deterministically from the ground-truth label, the smoothing strength α, and the prior π. The construction is therefore fully reproducible from a single label and does not require annotator votes.

Author Action: We have revised Section 3.1.1 to state explicitly that the benchmarks are single-label and that Equation (2) reduces to smoothed soft targets, and we have released the label-construction code in `fleo/fuzzy_loss.py` (activated by the `--fuzzy-alpha` training flag) so the fuzzy targets are reproducible.

Comment 1.3: What is the justification for the values of smoothing parameter α and prior distribution π? Please elaborate the construction pipeline of fuzzy labels and the rationale for parameter selection.

Author Response: We thank the reviewer for the comment. The construction pipeline is: (i) take the one-hot ground-truth label; (ii) leak a fraction α of the probability mass onto the remaining classes in proportion to the prior π; (iii) renormalise. We set α = [0.1], chosen by a small sweep α ∈ {0.05, 0.1, 0.2} on the RAF-DB validation split ([__] justifies 0.1), and π = [the empirical class-frequency prior], so probability leaks toward frequently-confused, well-populated neighbours while confident classes stay near one-hot. Larger α over-smooths minority classes; α = 0 recovers hard labels.

Author Action: We have revised Section 3.1.1 to describe the three-step construction pipeline and to report and justify α and π, and we have added the α sweep to the supplementary results.

Comment 1.4: Please ensure all mathematical notations in the equations are clearly defined immediately below the equations.

Author Response: We thank the reviewer for the comment. We agree that every symbol should be defined at its first use.

Author Action: We have added, immediately below each equation, a definition of every symbol (e.g. K, d, μ_c, n_c, α, π, λ_o, λ_b, Û, p, b) in Sections 3.1.1–3.1.3.

Comment 1.5: The English writing needs to improve, and the references need to be properly formatted.

Author Response: We thank the reviewer for the comment.

Author Action: We have carried out a full language edit of the manuscript for grammar and clarity, and we have reformatted all references to the MDPI Algorithms style, ensuring consistency and complete metadata.

Reviewer 2

Comment 2.1: The title should be reduced so that the number of lines of the reduced title is less than or equal to two.

Author Response: We thank the reviewer for the comment.

Author Action: We have shortened the title to "DPU-Native YOLOv8-FLEO: Fold-Out Orthogonalization for Real-Time FPGA Facial-Expression Recognition", which fits within two lines.

Comment 2.2: Delete the paragraph "The remainder of this paper … concludes the study." (lines 100–103), since its contents are redundant.

Author Response: We thank the reviewer for the comment and agree it is redundant.

Author Action: We have deleted the paper-organization paragraph at the end of Section 1.

Comment 2.3: Unify the expression of the same terms (e.g. "YOLOv8-FLEO framework", "YOLOv8-FLEO module", "YOLOv8-FLEO model").

Author Response: We thank the reviewer for the comment.

Author Action: We have unified the terminology to "YOLOv8-FLEO framework" for the overall system and "FLEO block" for the neck module throughout the manuscript.

Comment 2.4: Check whether the variables X and Y in Figures 1, 2, 3, and 4 are the same; use different symbols if some differ.

Author Response: We thank the reviewer for the comment.

Author Action: We have audited the axis/variable symbols across Figures 1–4 and disambiguated them, using distinct symbols where the quantities differ and a consistent symbol where they are identical.

Comment 2.5: Redraw Figure 1 (delete "High-Level System Architecture", "YOLO v8" top/bottom, "Proposed", "Block", "Panel B:", "Block Logic (Detailed)"; add the model name in bold at the top of Fig 1(A); remove training flows from 1(B)/1(C); reconcile the Fold-out Graph across panels; remove duplicate items in 1(C)/1(D); use vector graphics; enlarge small text; mention Output in 1(A)).

Author Response: We thank the reviewer for the detailed comment.

Author Action: We have redrawn Figure 1 as a vector graphic: removed the listed labels, added the model name in bold at the top of panel (A), added the Output stage, removed the training flows from the architecture panels, made the fold-out graph consistent across panels, removed the duplicated elements between panels (C) and (D), and enlarged all text to near main-text size.

Comment 2.6: Remove duplicate words in Figure 2, separate the FLEO Block subfigure, adjust text size/clarity, and merge Figure 2 into Figure 1.

Author Response: We thank the reviewer for the comment.

Author Action: We have merged Figure 2 into Figure 1 as an additional panel, removed the duplicated text, and re-typeset the FLEO block schematic as a clear vector subfigure with enlarged labels.

Comment 2.7: Redraw Figures 3 and 5 as academic (rather than engineering) figures: add more imagery, remove illustrative text, and do not distinguish box types by colour alone.

Author Response: We thank the reviewer for the comment.

Author Action: We have redrawn Figures 3 and 5 with representative imagery, reduced the explanatory text, and added shape/pattern encodings in addition to colour so the box types remain distinguishable without relying on colour.

Comment 2.8: Remove non-essential text from the "Value" columns of Tables 2 and 3 (e.g. replace "SGD (Momentum = 0.937)" with "SGD").

Author Response: We thank the reviewer for the comment.

Author Action: We have trimmed the "Value" columns of Tables 2 and 3 to concise entries, moving the detailed settings into the surrounding text.

Comment 2.9: Open a new section for evaluation metrics; add a heatmap figure; and compare the proposed model with two more models such as YOLO11 and YOLO26.

Author Response: We thank the reviewer for the comment. We agree that a dedicated metrics section and stronger baselines improve the evaluation.

Author Action: We have added a new "Evaluation Metrics" subsection defining every metric, added a confusion-matrix heatmap figure, and added YOLO11-FLEO and YOLO26-FLEO comparisons on RAF-DB in Table [X] ([__]).

Comment 2.10: Check spelling throughout (e.g. "bottleneckssuch" → "bottlenecks such", "recurrenceand" → "recurrence and", "workarounds").

Author Response: We thank the reviewer for the comment.

Author Action: We have corrected the noted spelling/spacing errors and run a full spell-check over the manuscript.

Comment 2.11: Check whether the affiliations of the first and third authors are correct, since the first half is the same while the second half differs; separate combined affiliations if needed.

Author Response: We thank the reviewer for the comment.

Author Action: We have verified and, where two institutions were combined into one entry, separated them into distinct numbered affiliations for the first and third authors.

Reviewer 3

Comment 3.1: Equation (2) derives fuzzy membership values from annotator vote counts n_c, yet the standard RAF-DB and FER2013 datasets provide a single label. Where do the per-image vote counts come from, and how are the seven-dimensional fuzzy targets generated? This is central to the "Fuzzy Label" contribution and should be fully reproducible.

Author Response: We thank the reviewer for the comment. As also clarified for Reviewer 1, the datasets are single-label and no per-image vote metadata is used. With a single label, $n_c = \mathbb{1}[c = y]$, so Equation (2) reduces to the deterministic smoothed target $\mu_c = (\mathbb{1}[c=y] + \alpha\cdot\pi_c)/(1 + \alpha)$. The seven-dimensional target is thus generated algorithmically from the label, α, and π, and is fully reproducible.

Author Action: We have revised Section 3.1.1 accordingly and released the generating code in `fleo/fuzzy_loss.py` (flag `--fuzzy-alpha`), so any reader can regenerate the fuzzy targets exactly.

Comment 3.2: Section 3.1.3 states that Gram–Schmidt is only a training-time regularizer that can be folded out at inference, yet Table 7 shows that simply removing it collapses RAF-DB accuracy from 0.858 to 0.073. Please clarify the distinction between a training-only regularizer and a forward operator that remains essential until the network is fine-tuned without it.

Author Response: We thank the reviewer for this important observation, which is correct. Gram–Schmidt is not a no-op at inference; naively deleting it does break the forward path (0.858 → 0.073). Our claim is a two-step deployment: (i) fold-out removes the DPU-hostile Gram–Schmidt operator, and (ii) a short post-fold fine-tuning (10 epochs, no orthogonality) lets the remaining DPU-native convolutions re-absorb the function Gram–Schmidt was performing (recovering to 0.849). The benefit is therefore internalized into the weights by fine-tuning, not "already free". We have corrected the wording to avoid describing Gram–Schmidt as a pure training-time regularizer.

Author Action: We have rewritten Section 3.1.3 and the abstract to describe the mechanism as "fold-out followed by post-fold fine-tuning", and we have annotated Table 7 to make the two-step recovery explicit.

Comment 3.3: The authors can refer to some current works, such as: (1) Visually steered reconfigurable intelligent surface-assisted mobile communications; (2) Metasurface Router for Near-Field Multi-User Low-Interference Access Driven by Structured Wave; (3) Azimuth-Dependent Secure Transmission With Orbital Angular Momentum Directional Modulation; (4) Techniques of 2D Human Pose Estimation Based on Computer Vision: A Survey.

Author Response: We thank the reviewer for the suggested references.

Author Action: We have added the four suggested works to the references and cited them in the related-work discussion of reconfigurable/edge systems and computer-vision techniques.

Comment 3.4: Ten additional fine-tuning epochs recover accuracy from 0.073 to 0.849. How much would a standard YOLOv8s gain from the same 10 additional epochs and learning-rate schedule? Without this control, part of the recovered performance may come from extra optimization rather than from information internalized through orthogonalization.

Author Response: We thank the reviewer for this valuable point. We have added the requested control: a standard YOLOv8n trained for the same 10 additional epochs under the identical schedule ($\eta_0 = 5\times10^{-4}$ cosine, no orthogonality). The control gains [**] mAP50, whereas the fold-out + fine-tuned FLEO reaches [**]; the gap of [__] points is attributable to the orthogonalization internalized during training rather than to extra optimization alone.

Author Action: We have added this control to Table 7 (or a new Table [X]) and discussed it in Section 4, so the recovery is properly attributed.

Sincerely,

The Authors